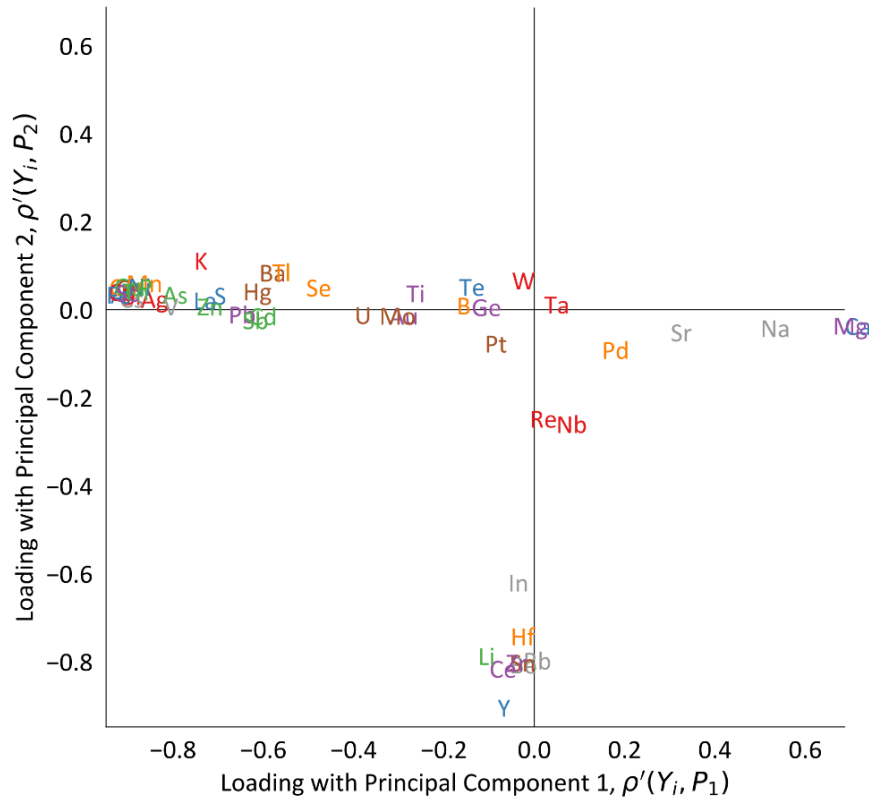


Lecture 14 – Reducing Dimensionality I



Today's Learning Outcomes

1. Know what the purpose of a principal component analysis (PCA) is
2. Be able pre-process and run a PCA in Python
3. Know how to visualize results to identify meaningful principal component axes and interpret them

Multivariate Data

It is very common to have datasets with many correlated variables in geology

- Ex. geochemical analyses

Variables are often strongly correlated and thus not independent

- Ex. Ca and Mg concentrations tend to positively correlate with each other and negatively correlate with Zr or Ti

Multivariate Data

In those cases where we have correlated explanatory variables which variable do I use to understand a system???

- None are perfectly correlated, so they each have some differing amounts of signal and error

Could run a stepwise regression analysis

- Find the variable that explains the most data, then controlling for that find the variable which explains the next most variance and then repeat (forward analysis)

Stepwise Regression Procedure

$$y \sim c \rightarrow AIC = 560$$

$$y \sim x_1 + c \rightarrow AIC = 400$$

$$y \sim x_1 + x_2 + c \rightarrow AIC = 380$$

$$y \sim x_1 + x_2 + x_3 + c \rightarrow AIC = 375$$

$$y \sim x_1 + x_2 + x_3 + x_4 + c \rightarrow AIC = 378$$

Forward approach



Favored model by addition

$$y \sim x_1 + x_2 + x_3 + x_4 + \dots + x_n + c \rightarrow AIC = 442$$

$$y \sim x_1 + x_2 + x_3 + x_4 + x_5 + c \rightarrow AIC = 400$$

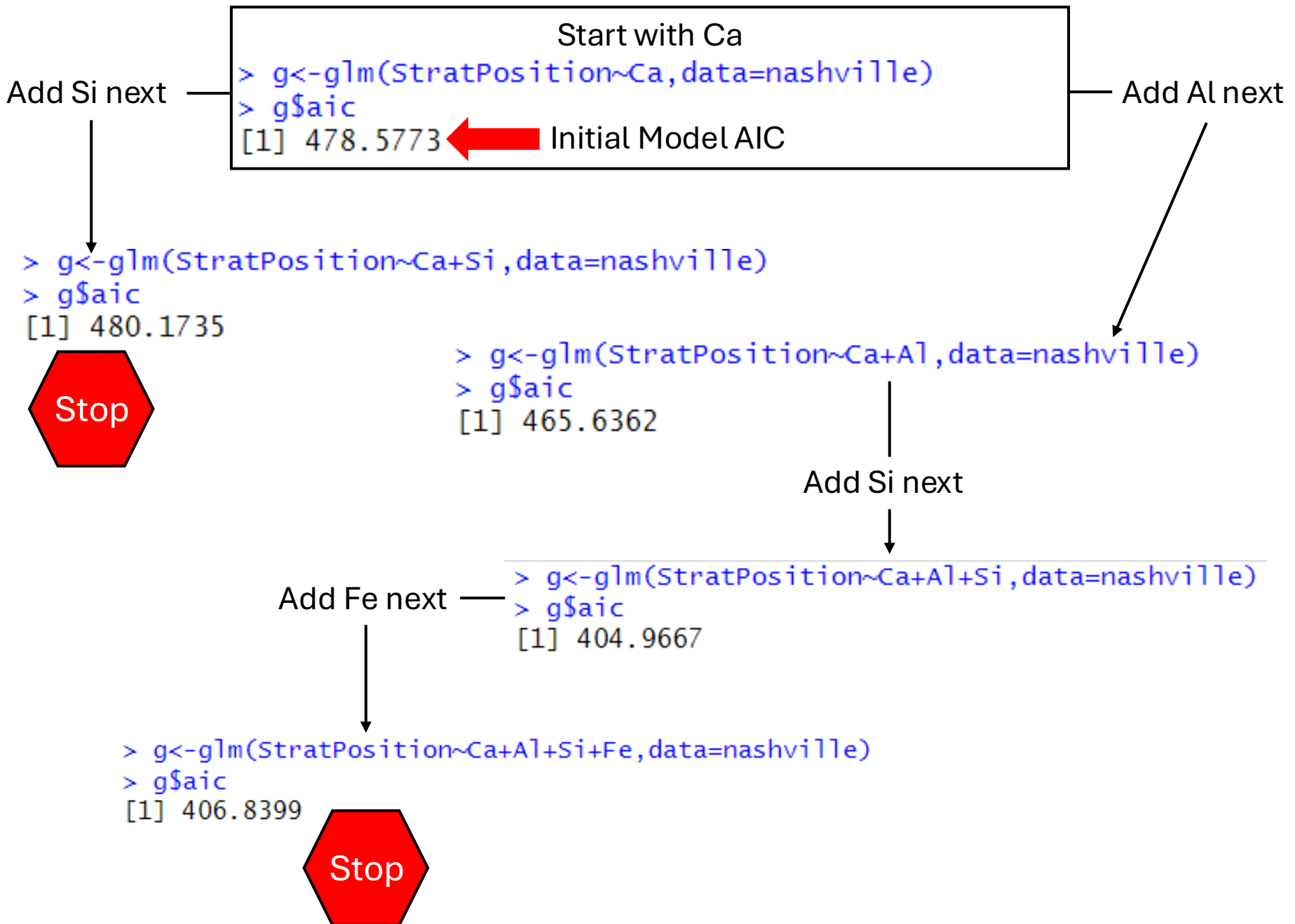
$$y \sim x_1 + x_2 + x_3 + x_4 + c \rightarrow AIC = 378$$

$$y \sim x_1 + x_2 + x_3 + c \rightarrow AIC = 375$$

Backward approach



Favored model by subtraction



Stepwise Regression Cautions

For n variables, how many experiments do we need to do?

- Stepwise regression

$$n + (n - 1) + \cdots + 1 = \frac{n(n + 1)}{2}$$

- All subsets (Best subset regression)
 2^n

Stepwise regression tries to avoid evaluating all subsets, but its outcome depends on the order in which variables are entered.

That's why stepwise regression is unstable with correlated variables.

If some variables are strongly correlated whichever gets added first added or last deleted will be the one that shows up even though any would have been equally as good

- Different orders will give different answers!

Stepwise Regression cannot address the correlated variable challenge...

The Core Problem: Too Many Correlated Variables

- Many geochemical variables move together (e.g., Ca–Mg, Al–Fe–Si).

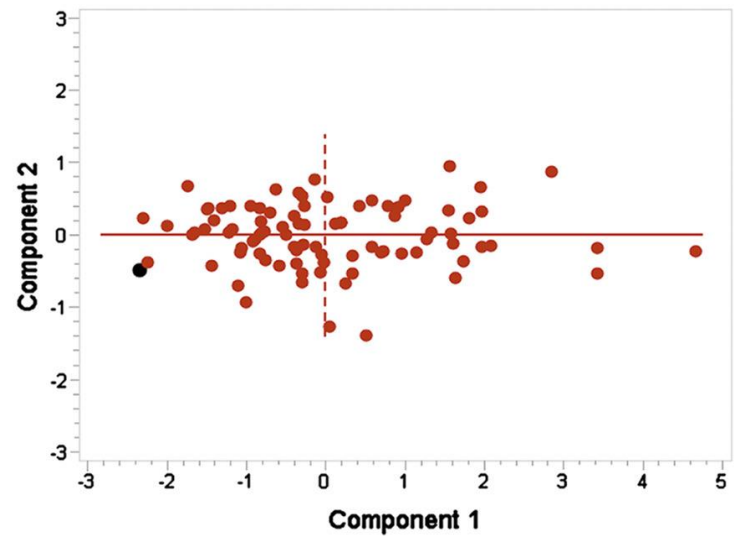
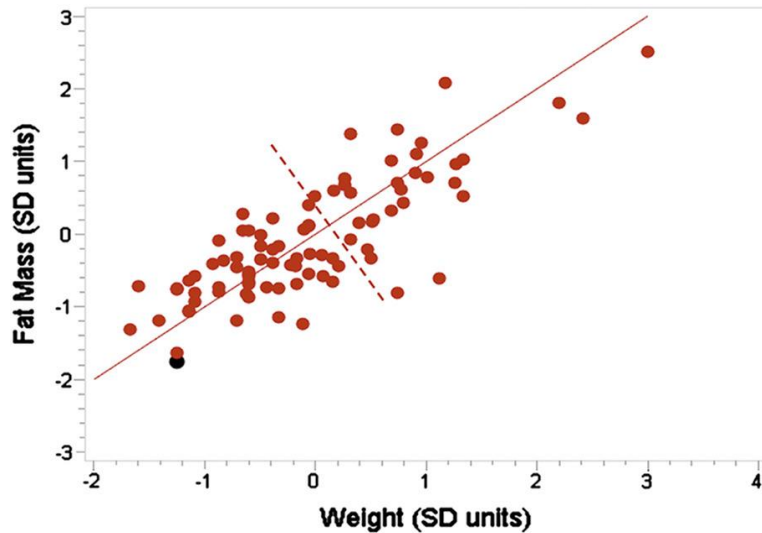
Correlated variables create problems:

- hard to visualize multivariate patterns
- hard to interpret individual effects
- hard to build stable predictive models

We need a method to summarize correlated variables into a smaller number of meaningful axes.

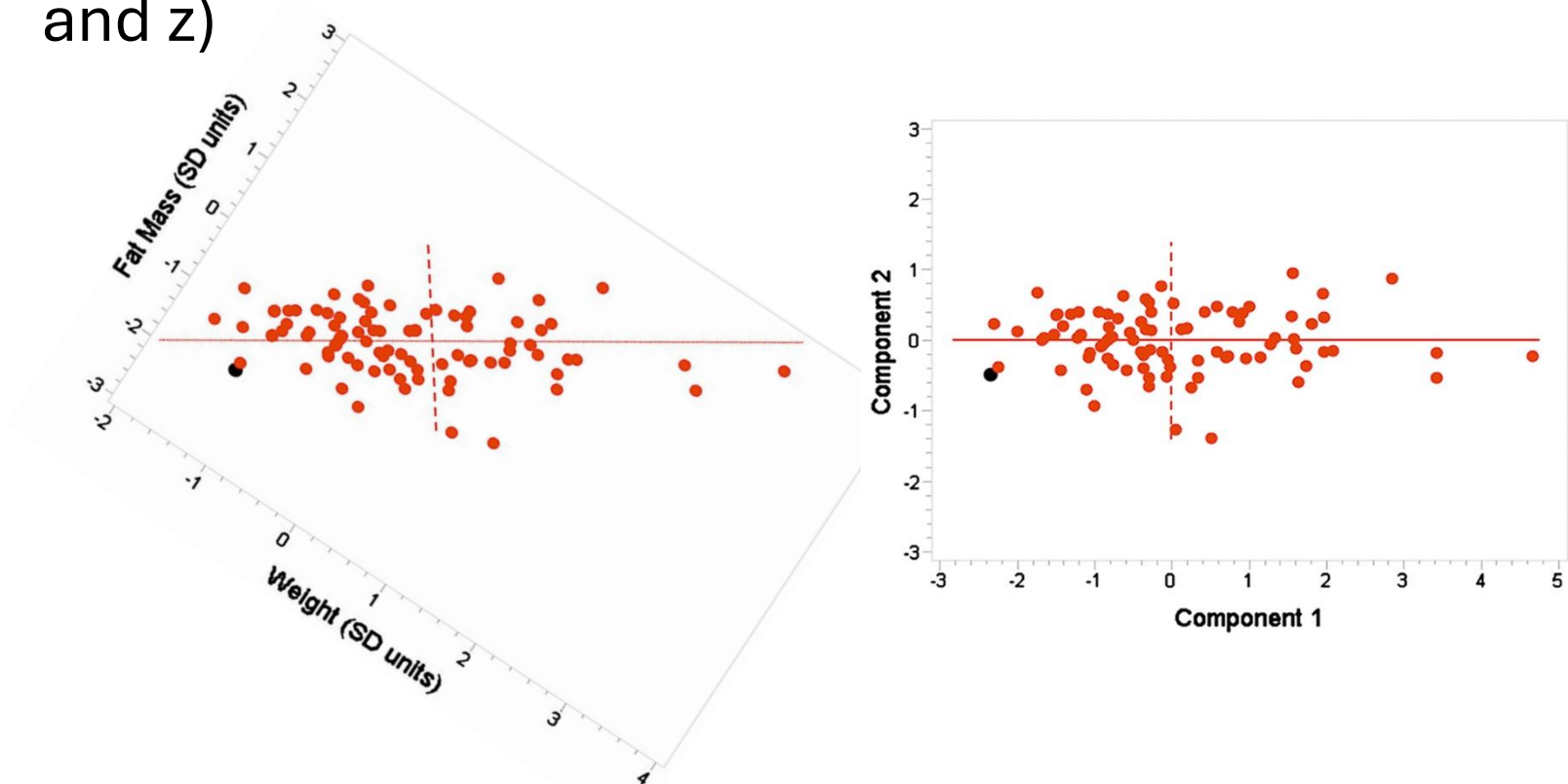
Principal Component Analysis (PCA)

Unlike stepwise regression, **PCA** can transform the original variables to create new composite variables which are independent of one another



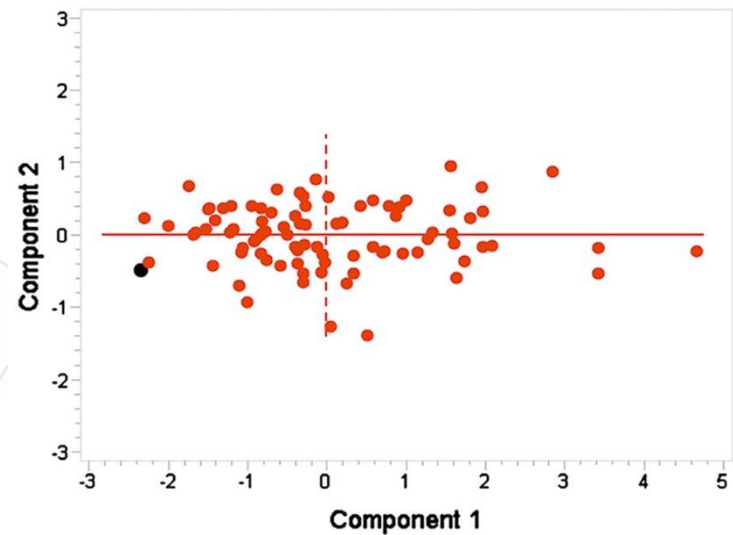
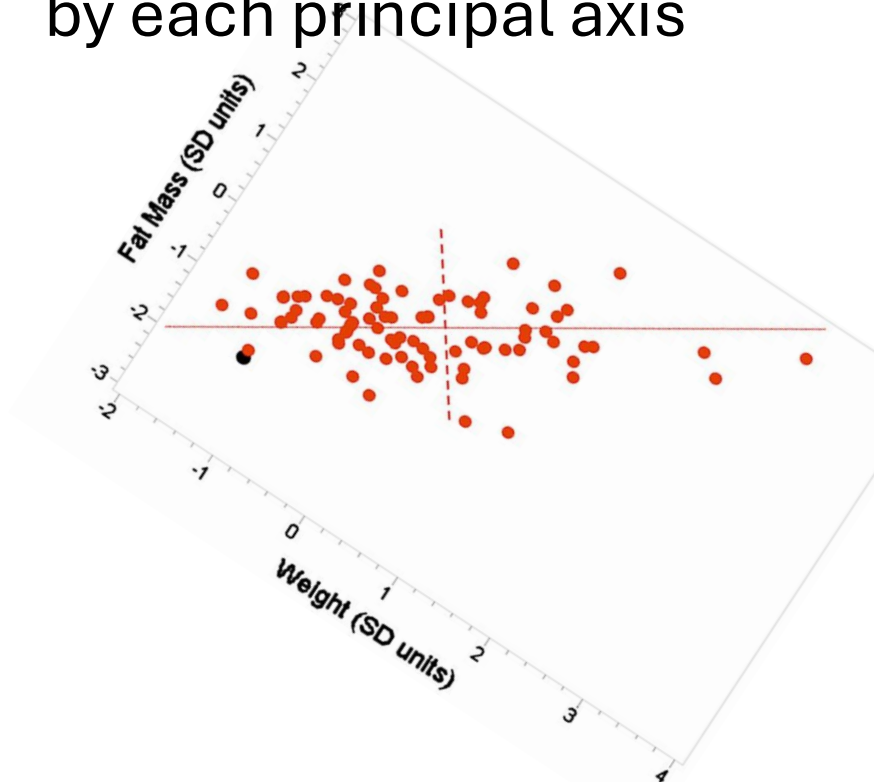
Principal Component Analysis (PCA)

Takes data and re-centers it with axes aligned to the greatest variation in the data, and the orthogonal axes (in math you can have infinite dimensions, not just x, y and z)



Principal Component Analysis (PCA)

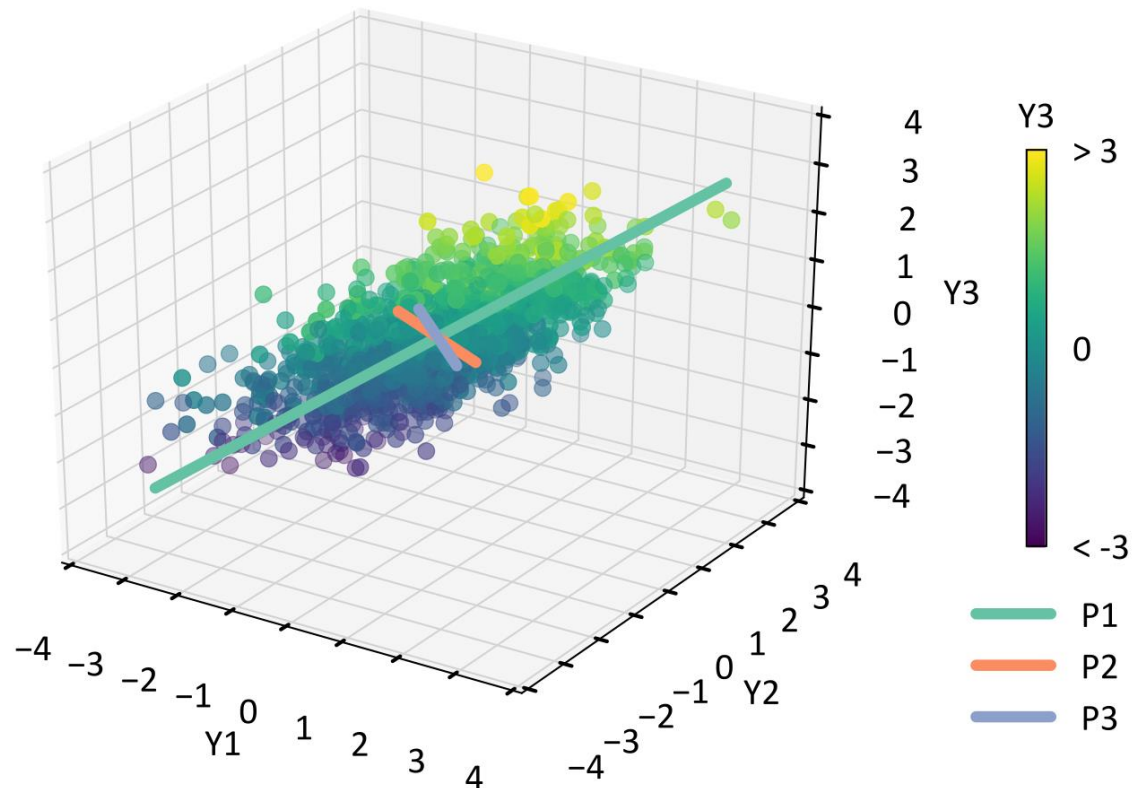
New coordinates in transformed space calculated and from this a set correlation matrices are formed which gives a percent variance of the original data explained by each principal axis



3D Example Visualization

Example Plot of original data with 3 variables (Y1, Y2, Y3)

P1, P2, and P3 are the principal component axes where the length of the lines are their proportional variance explained in the data



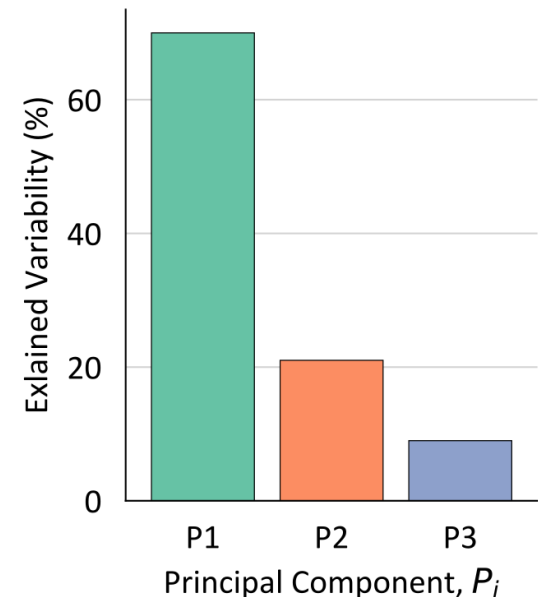
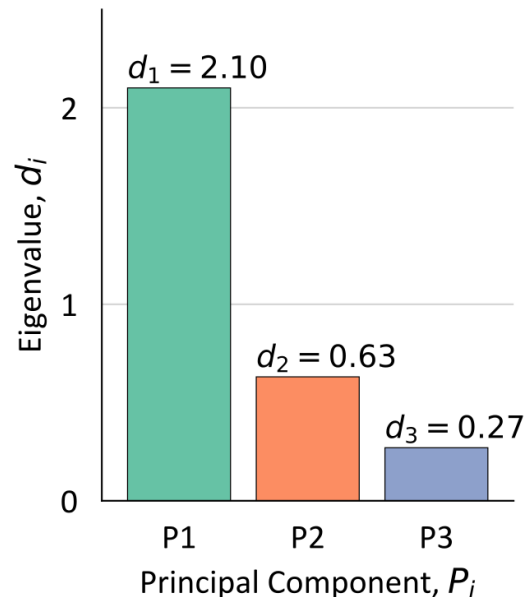
Variance Explained

By definition the principal component axes will go from most variance explained to least variance explained

- Variance explained is extracted from the transformation matrix in PCA analyses

Here we have 3 variables,
so total variance = 3

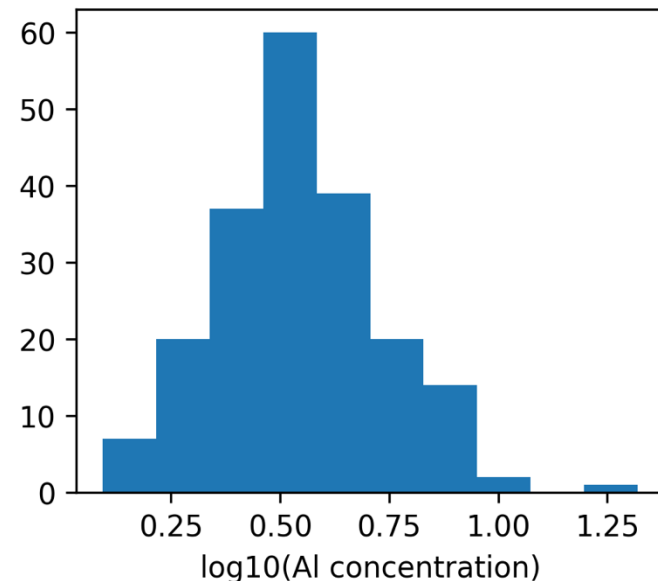
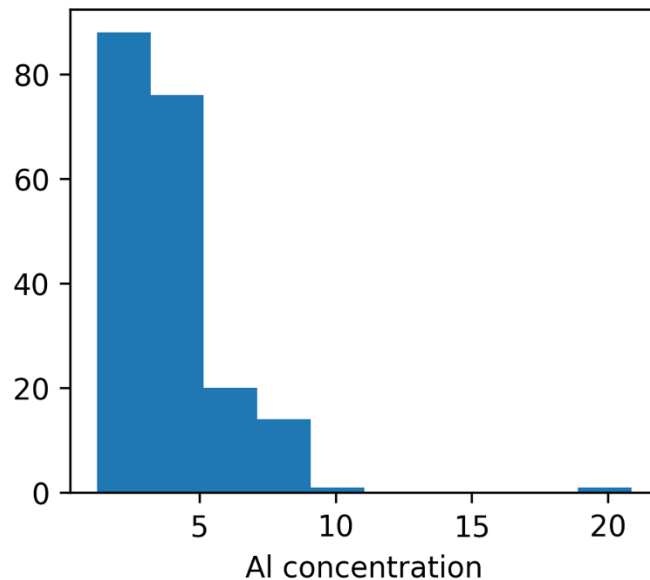
% Var. exp. of *axis 1* =
 $2.1 / (2.1 + 0.63 + 0.27) = 70\%$



Necessary data preparation for applying PCA in geological data

Log-Transform is necessary for skewed dataset (first step)

- Geochemical Data Are Often Right-Skewed
- Concentrations vary over orders of magnitude
- Log-transform can reduce skewness, makes distribution more symmetric, and stabilizes variance



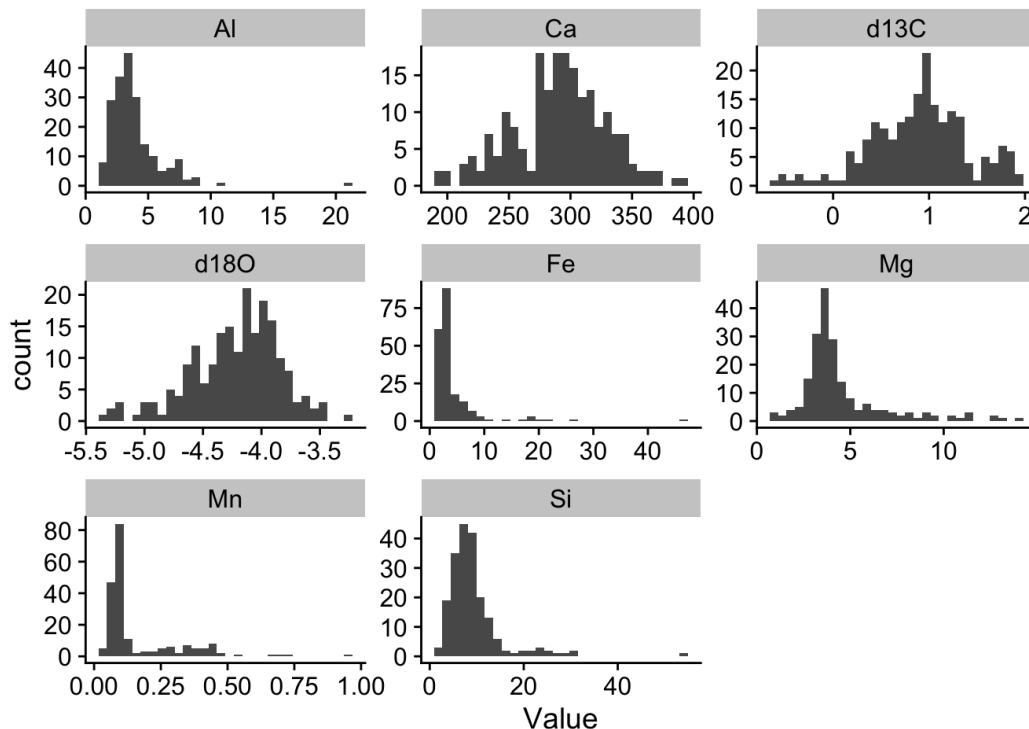
Necessary data preparation for applying PCA in geological data

Normalize data because Different Units & Scales = Biased PCA.

- For example, $\delta^{13}\text{C}$ ranges from ~ -1 to $+2$, Ca ranges from ~ 200 to ~ 400 , Mn spans orders of magnitude
- Raw data would make the “biggest-scale variable” dominate PC1.
- **Normalization gives each variable equal opportunity to contribute.**

Example – Nashville Carbonate Dataset

This data set has geochemical analyses of 200 limestone samples, including the major elements Al, Ca, Fe, K, Mg, Mn, and Si, stable isotope ratios of carbon and oxygen (d13C and d18O). It also records the stratigraphic position of the samples.



PCA in Python

PCA analysis in Python

```
from sklearn.decomposition import PCA
```

```
# Log10-transform
```

```
cols_to_log = ['Al', 'Ca', 'Fe', 'Mg', 'Mn', 'Si']
```

```
geochem[cols_to_log] = np.log10(geochem[cols_to_log])
```

```
# Normalize
```

```
scaler = StandardScaler()
```

```
X_scaled = scaler.fit_transform(geochem.values)
```

```
# PCA
```

```
pca = PCA() # keep all components
```

```
scores = pca.fit_transform(X_scaled)
```

Expected output

As you might see from previous slides, it is a breeze to apply PCA in Python.

What are the expected outputs and how can we interpret those outputs?

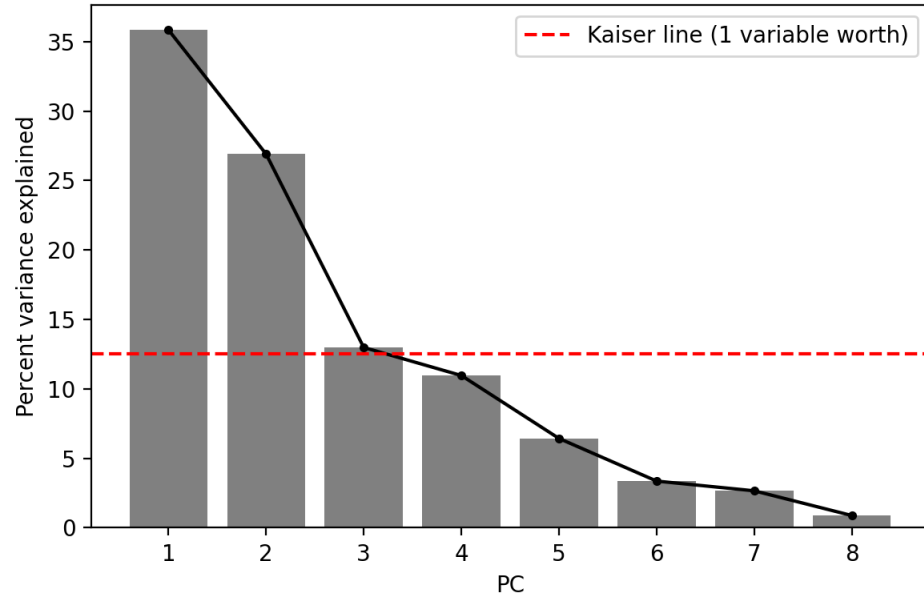
1. Variance (or Eigenvalues)
2. Loadings
3. Correlation loadings
4. Scores

The importance of PC axis is quantified using variance

In most systems the first few axes will explain the vast majority of the variance

Cutoff for how many axes to consider are determined by...

- % var. exp. > 1 variable's worth
- Cumulative to 95% var. exp.
- Permutation test



PCA Loadings

Loadings ($a_{i,j}$) are the **weights** used to construct each PC ($j = 1, \dots, 8$) as a linear combination of variables ($i = 1, \dots, 8$).

$$PC_j = \sum_{i=1, \dots, n}^n a_{i,j} X_i$$

	PC1	PC2	PC3	PC4	PC5	PC6	PC7	PC8
d13C	0.17	-0.23	0.64	-0.57	-0.42	0.06	0.06	0.00
d18O	-0.05	0.60	0.06	-0.26	0.24	0.55	0.45	-0.04
Al	0.50	0.29	0.03	0.18	-0.16	0.05	-0.15	0.76
Ca	-0.33	0.04	0.47	0.65	-0.26	-0.03	0.40	0.05
Fe	0.53	-0.09	-0.10	-0.01	0.16	-0.47	0.66	-0.12
Mg	0.06	0.55	0.44	0.00	0.27	-0.48	-0.36	-0.26
Mn	0.23	-0.42	0.39	0.21	0.67	0.33	-0.10	0.04
Si	0.51	0.09	-0.06	0.32	-0.36	0.35	-0.17	-0.58

PCA Loadings

Loadings ($a_{i,j}$) are the **weights** used to construct each PC ($j = 1, \dots, 8$) as a linear combination of variables ($i = 1, \dots, 8$).

- Loadings tell you the direction of each PC, but do NOT tell you how strongly each original variable correlates with that PC.

	PC1	PC2	PC3	PC4	PC5	PC6	PC7	PC8
d13C	0.17	-0.23	0.64	-0.57	-0.42	0.06	0.06	0.00
d18O	-0.05	0.60	0.06	-0.26	0.24	0.55	0.45	-0.04
Al	0.50	0.29	0.03	0.18	-0.16	0.05	-0.15	0.76
Ca	-0.33	0.04	0.47	0.65	-0.26	-0.03	0.40	0.05
Fe	0.53	-0.09	-0.10	-0.01	0.16	-0.47	0.66	-0.12
Mg	0.06	0.55	0.44	0.00	0.27	-0.48	-0.36	-0.26
Mn	0.23	-0.42	0.39	0.21	0.67	0.33	-0.10	0.04
Si	0.51	0.09	-0.06	0.32	-0.36	0.35	-0.17	-0.58

PAC correlation loadings

Correlation Loadings explains “How strongly each variable correlates with each PC”

PCA loadings scaled by the square root of eigenvalue

$$\text{Correlation Loading}_{i,j} = a_{i,j} \times \sqrt{\text{eigenvalue}_j}$$

Difference between loading and correlation loading.

- Loadings describe the direction of a variable in PC space.
- Correlation loadings describe the strength of the variable's correlation with each PC.

	PC1	PC2	PC3	PC4	PC5	PC6	PC7	PC8
d18O Loading	-0.05	0.60	0.06	-0.26	0.24	0.55	0.45	-0.04
d18O Correlation Loading	-0.08	0.88	0.06	-0.24	0.17	0.29	0.21	-0.01

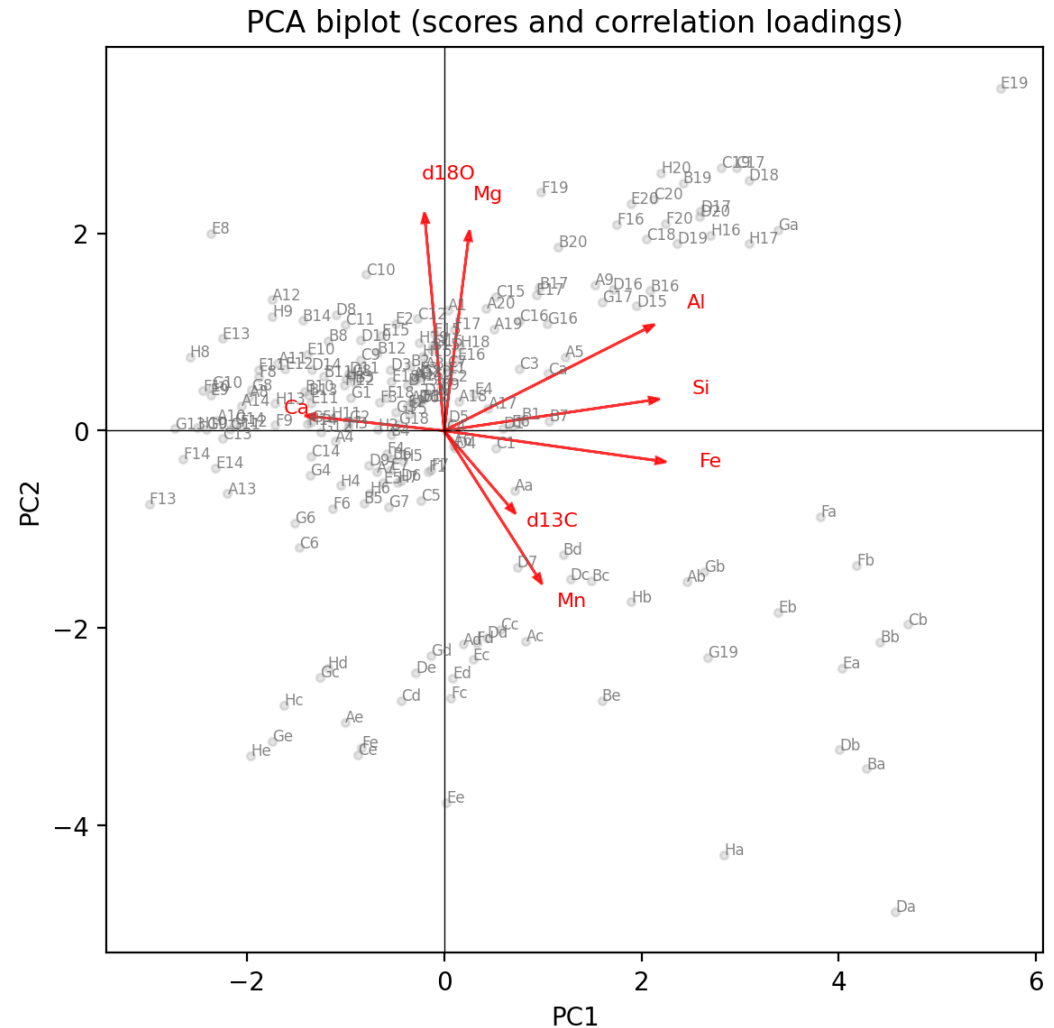
PCA Biplot

Red arrows show the correlation loadings of each variable as a vector for PC1 and PC2

Ex. Ca has a negative positive loading for PC1, but a near-zero loading for PC2

Scores of each sample (grey labels) for PC1 and PC2

- **Scores** tell where the *samples* are on the new axes.



Summary of PCA Workflow

1. Examine data to remove or minimize the impact of outliers (log-transform)
2. Normalize data (z-transform where mean = 0, variance = 1)
3. Run PCA
4. Extract the variance explained by each principal component
5. Test how many axes are significant or meaningful
6. Extract loadings, correlation loadings, and scores to visualize (biplot) and interpret the meaning of the principal component axes

Today's Learning Outcomes

1. Know what the purpose of a principal component analysis (PCA) is
2. Be able pre-process and run a PCA in R using the `prcomp()` function
3. Know how to visualize results to identify meaningful principal component axes and interpret them